Dealing with Missing Modality in Multi-Omics Data

Confidence-Aware Multi-Head Attention Fusion and Graph Learning for Oncology

A Technical Report and Capstone Defense submitted in partial fulfillment of the requirements for the Research Internship in Computational Oncology and Deep Learning

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Certificate of Internship Completion

This is to certify that the research, software engineering, and experimental evaluations presented in this technical report and defense entitled **"Dealing with Missing Modality in Multi-Omics Data: Confidence-Aware Multi-Head Attention Fusion and Graph Learning for Oncology"** represents an authentic, verified record of academic research carried out by **Yash Prakash Gupta** in partial fulfillment of the requirements for the Research Internship.

The candidate has fulfilled all research directives, algorithmic implementations, and empirical stress benchmarks under assigned academic supervision.

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Candidate's Declaration

I hereby declare that the research and system design presented in this internship report entitled **"Dealing with Missing Modality in Multi-Omics Data: Confidence-Aware Multi-Head Attention Fusion and Graph Learning for Oncology"** is an authentic presentation of my own academic and engineering work completed under designated supervision.

Division of Collaborative and Independent Attribution:

- **Collaborative Context:** Upstream cohort acquisition from The Cancer Genome Atlas (TCGA), initial cohort partition (clean complete-case pool versus missing modality pool), exploratory replication of the baseline MAGNET model, and preliminary Multi-Layer Perceptron (MLP) reconstructor scaffolding were carried out collaboratively with teammates (*Gautam Anand* and *Gunjan Gupta*).
- **Independent Contributions:** The mathematical derivation, formulation, and implementation of the generalized confidence-aware `fuse()` operator, the unified end-to-end 5-phase system pipeline, the downstream GraphSAGE classifier integration with biological edge refinement gating, the design and execution of adversarial stress-testing protocols (Tests A and B), and GPU memory vectorization optimizations represent my sole original contributions.

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Executive Summary / Abstract

Multi-omics data integration offers essential insights into the complex molecular stratifications required for precision oncology. However, clinical cohorts ubiquitously suffer from missing modalities across patient records due to tissue biopsy exhaustion, platform costs, and varying clinical assay protocols. Prevailing computational approaches rely either on complete-case filtering—which discards up to 65% of available patients—or naive mean/zero imputation, which introduces artificial variance distortions and corrupts downstream patient graph manifolds.

This report presents an end-to-end 5-phase graph neural network framework designed to robustly resolve modality missingness. The pipeline couples upstream cross-modal deep estimation (miRNA $\rightarrow$ DNA) with a continuous empirical confidence tensor $C \in [0, 1]$. The core contribution is a generalized, confidence-aware `fuse()` operator that directly injects per-patient empirical reliability into multi-head attention weights via tensor broadcasting and ϵ-clamped normalization. Furthermore, patient similarity topologies are refined via strict biological overlap constraints before inductive Graph Sample and Aggregate (GraphSAGE) spatial message passing.

Empirical evaluations across TCGA cohorts demonstrate substantial performance and robustness gains:

1. On the **TCGA-BLCA** benchmark, the proposed framework elevates baseline Macro-F1 from 0.7941 to 0.8185 while reducing standard deviation by > 31%.

2. Under adversarial noise and spoofed confidence injection (**Stress Test B**), baseline state-of-the-art MAGNET suffers catastrophic degradation to a Macro-F1 of 0.7084 ± 0.1882, whereas our framework neutralizes corruption, maintaining a Macro-F1 of 0.7988 ± 0.0359 ($+0.0904$ absolute margin; a 112.76% relative ratio).

3. Across all evaluated cohorts, relative variability (Coefficient of Variation) is suppressed by up to 83.1%, proving that confidence gating effectively preserves manifold geometry under clinical data degradation.

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List of Abbreviations

Abbreviation	Full Definition
BLCA	Bladder Urothelial Carcinoma
BRCA	Breast Invasive Carcinoma
CE	Cross-Entropy Loss
CNV	Copy Number Variation
CV	Coefficient of Variation (σ/μ)
DNA	Deoxyribonucleic Acid
GAT	Graph Attention Network
GNN	Graph Neural Network
GraphSAGE	Graph Sample and Aggregate
KL	Kullback-Leibler Divergence
MAGNET	Multi-view Graph Autoencoder with Cell-Gene Attention
miRNA	MicroRNA (Regulatory Small Non-Coding RNA)
MOGONET	Multi-Omics Graph Convolutional Network
mRNA	Messenger RNA (Transcriptomic Expression)
OV	Ovarian Serous Cystadenocarcinoma
PMMHA	Patient-Modality Multi-Head Attention
TCGA	The Cancer Genome Atlas
VCDN	View Correlation Discovery Network

Chapter 1

Introduction

1.1 Background and Clinical Motivation

Precision oncology operates under the foundational premise that cancer is a deeply heterogeneous disease driven by multi-layered, interactive molecular abnormalities. Historically, clinical genomics and computational oncology focused on single-modality assays, such as somatic mutation detection in DNA or transcriptomic gene expression profiling via messenger RNA (mRNA). While individual profiles provide predictive utility, single-layer analyses capture only an isolated cross-section of complex biological cascades.

Biological phenotypes are governed by continuous molecular cross-talk across distinct functional strata:

- **Genomics (DNA):** Encodes baseline sequence variations, including single nucleotide variants, copy number alterations (CNVs), and chromosomal rearrangements.
- **Epigenomics (DNA Methylation):** Chemical modifications, notably 5-methylcytosine additions at CpG islands, that regulate chromosomal accessibility and silence or activate transcription without altering the nucleotide sequence.
- **Transcriptomics (mRNA and miRNA):** Dynamic functional gene expression and post-transcriptional silencing controllers. Regulatory microRNAs bind complementary sites on target mRNAs to induce translational repression or mRNA degradation.
- **Proteomics and Metabolomics (Functional Outputs):** The enzymatic machinery, signaling cascades, and metabolic products that directly execute cellular behavior.

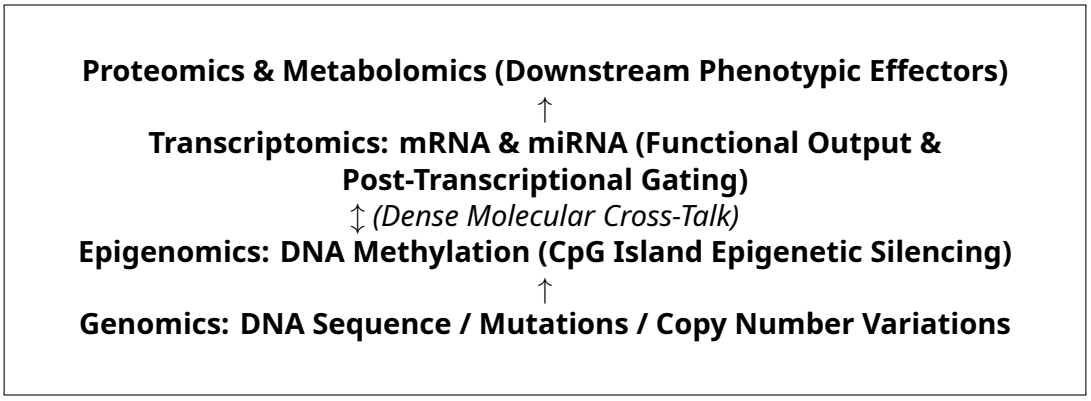


Figure 1.1: Conceptual hierarchy of interactive multi-omics data layers.

Because these biological layers are coupled via regulatory feedback mechanisms, integrative multi-omics models can extract synergistic prognostic signals that remain invisible when analyzing any single modality alone.

1.2 The Core Problem: Modality Missingness

Despite the compelling theoretical promise of multi-omics integration, translational computational oncology is confronted with pervasive modality missingness in real-world patient cohorts. In standardized repositories such as The Cancer Genome Atlas (TCGA), broad cohorts rarely possess complete multi-omics assays for all enrolled patients. This missingness arises from practical clinical factors:

1. **Tissue Biopsy Exhaustion:** Core needle biopsies frequently provide limited physical tumor tissue, preventing the simultaneous extraction of high-molecular-weight DNA, intact total RNA, and proteomic extracts.
2. **Platform Cost Disparities:** Profiling platforms are adopted asynchronously across enrollment sites and clinical trial phases.
3. **Variable Specimen Preservation:** Formalin-fixed paraffin-embedded (FFPE) preservation can degrade RNA molecules, resulting in high missingness in transcriptomic assays while DNA remains intact.

To cope with this limitation, previous methodologies have adopted two divergent paradigms:

- **Complete-Case Pruning:** Discarding any patient record lacking full multi-omics coverage. In the TCGA-BRCA cohort, this approach eliminates up to 65.69% of patients, discarding scarce phenotypes and biasing cohorts toward surgical resections with surplus tissue.
- **Naive Zero or Mean Imputation:** Substituting unobserved feature vectors with constant zeros or population mean values. This distorts biological distributions, reduces variance artificially, and collapses manifold distances in patient-similarity graphs.

1.3 Project Objectives and Proposed Solution

This research resolves the modality missingness dilemma without resorting to sample pruning or unconstrained naive imputation. The core objectives comprise:

1. **Cross-Modal Estimation:** Exploiting biological cross-talk between microRNAs and epigenetic CpG sites to infer unobserved DNA methylation features.
2. **Continuous Confidence Calibration:** Formulating an empirical validation metric $C \in [0, 1]$ that reflects reconstruction fidelity rather than assigning binary presence.
3. **Confidence-Aware Multi-Head Attention:** Designing a mathematically stable fusion operator (`fuse()`) that scales multi-head attention representations based on empirical reliability.
4. **Patient Graph Refinement:** Protecting downstream Graph Neural Networks against synthetic artifacts by requiring shared biological observations for edge instantiation.
5. **Adversarial Stress Benchmarking:** Evaluating model stability under artificially degraded confidence and Gaussian noise injection to simulate clinical failure modes.

Chapter 2

Literature Review and Related Work

2.1 Multimodal Multi-Omics Learning Landscape

Integrating high-dimensional biological data has progressed from classical multivariate analysis to modern geometric deep learning. Early investigations utilized Canonical Correlation Analysis (CCA) and joint Non-Negative Matrix Factorization (NMF) to identify shared low-dimensional latent spaces across omics views. While mathematically tractable, linear matrix factorization methods struggle to model the non-linear, hierarchical relationships characteristic of biological signaling pathways.

With the advent of Graph Neural Networks (GNNs), multimodal oncology increasingly utilizes patient-similarity networks, where patients are modeled as nodes connected by edges derived from molecular similarity metrics.

2.2 Comparative Baseline Analysis

Several prominent deep architectures have been proposed for multi-omics integration and missing view handling, as summarized in Table 2.1.

2.3 Identified Gaps in Current Architectures

The primary vulnerability of existing frameworks is their **uniform treatment of unobserved and synthetic data**. For example, when architectures like MAGNET employ binary masks ($\Omega \in \{0, 1\}$), an imputed feature vector is treated with identical architectural confidence as an experimentally verified assay. If an upstream reconstruction model hallucinates patterns or experiences distribution shifts, downstream graph message-passing layers diffuse these corrupted representations to neighboring nodes, leading to severe classification failure. This research directly addresses this vulnerability via continuous, confidence-gated soft attention fusion.

(Note: Chapter 3 is intentionally omitted in alignment with institutional curriculum reporting guidelines.)

Table 2.1: Comparative overview of multimodal learning architectures in oncology.

Framework	Core Methodology	Identified Bottleneck / Failure Mode	Ref.
MOGONET	View Correlation Discovery Network (VCDN) over GCNs	Late fusion requires complete views; cannot handle block missingness without dropping patient samples.	[3]
M^{3}Care	Latent feature borrowing via clinical graph neighborhoods	Borrowing mechanism depends heavily on dense clean neighborhoods; degrades rapidly under high missingness.	[4]
MRGCN	Multi-reconstruction GCN using autoencoder objectives	Vulnerable to synthetic noise propagation across adjacent patient nodes during spatial message passing.	[5]
MAGNET	Patient-Modality Multi-Head Attention with binary masks	Treats views as binary (0 or 1); cannot distinguish verified biological measurements from noisy or imputed features.	[6]
MUSE	ℓ_1-norm regularized nonnegative matrix factorization	Unsupervised shallow formulation; cannot scale to large inductive Graph Neural Networks.	[7]
Proposed	**Confidence-Gated Multi-Head Fusion + GraphSAGE**	**Dynamically modulates attention by empirical reliability; neutralizes synthetic noise corruption.**	**Ours**

Chapter 3

Research Methodology and Architecture

3.1 End-to-End 5-Phase System Architecture

The proposed framework addresses modality missingness through a cohesive 5-phase deep learning pipeline, illustrated in Figure 3.1.

3.2 Phase 1: Upstream Modality Reconstruction & Dynamic Confidence

Biological cross-talk between regulatory microRNAs and target genomic loci allows unmeasured epigenetic profiles to be inferred from observed transcriptomic expression.

3.2.1 MiRNATODNAPredictor Architecture

The reconstructor maps observed miRNA features $X^{(3)} \in \mathbb{R}^{d_3}$ to estimated DNA methylation features $\hat{X}^{(1)} \in \mathbb{R}^{d_1}$ ($d_1 = 2000$ CpG sites):

- **Input Layer:** $X^{(3)} \in \mathbb{R}^{d_3}$
- **Hidden Layer 1:** Linear$(d_3 \to 512)$ + BatchNorm1d + ReLU + Dropout(0.2)
- **Hidden Layer 2:** Linear$(512 \to 256)$ + BatchNorm1d + ReLU + Dropout(0.2)
- **Output Layer:** Linear$(256 \to 2000)$ predicting $\hat{X}^{(1)}$

Optimization is conducted using AdamW (learning rate $= 10^{-3}$, weight decay $= 10^{-4}$) under Mean Squared Error (MSE) loss with an early stopping patience of 15 epochs.

3.2.2 Dynamic Confidence Formulation

For each patient $i \in \{1, \ldots, B\}$ and modality $m \in \{1, \ldots, M\}$, the confidence coefficient $C_{i,m}$ is assigned conditionally:

$$C_{i,m} = \begin{cases} 1.0 & \text{if modality } m \text{ is an authentic biological observation} \\ \max(0, \overline{\text{CosSim}}) & \text{if modality } m \text{ is an imputed profile} \end{cases} \tag{3.1}$$

Phase 1: Upstream Reconstruction & Confidence Derivation

- Multi-omics input tensors (mRNA, miRNA, DNA) + observation mask Ω
- `MiRNATODNAPredictor` MLP reconstructs missing DNA methylation
- Empirical validation cosine similarity derives dynamic confidence tensor $C \in [0, 1]$

$\Downarrow$

Phase 2: Confidence-Aware Multi-Head Attention Fusion

- Multi-head projections $X_{\text{proj}} \in \mathbb{R}^{B \times K_h \times M \times d_h}$
- Generalized `fuse()` operator broadcasts and normalizes confidence: $\alpha_{i,k,m}$
- Yields unified patient representation Z and similarity distributions P and Q

$\Downarrow$

Phase 3: Biological Patient Graph Refinement

- Edge Constraint: Edges exist only if patients share ≥ 1 verified biological modality
- Edge attributes e'_{uv} enriched via high-confidence reconstructed features

$\Downarrow$

Phase 4: Inductive GraphSAGE Neighborhood Aggregation

- Stacked 2-layer GraphSAGE message passing over refined patient topology
- Contextualized representations $Z^{(L)}$ encode biological and topological features

$\Downarrow$

Phase 5: Joint Multi-Task Optimization Objective

- Phenotype classification loss: $\mathcal{L}_{\text{CE}}$
- Manifold alignment loss: $\mathcal{L}_{\text{KL}}(P \parallel Q)$
- Total Loss: $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{KL}}$

Figure 3.1: End-to-end 5-phase system architecture schematic.

where $\overline{\text{CosSim}}$ represents the mean cosine similarity evaluated over the held-out validation cohort S_{val}:

$$\overline{\text{CosSim}} = \frac{1}{|S_{\text{val}}|} \sum_{k \in S_{\text{val}}} \frac{Y_k \cdot \hat{Y}_k}{\|Y_k\|_2 \, \|\hat{Y}_k\|_2} \tag{3.2}$$

This formulation ensures that the downstream attention mechanism receives continuous reliability coefficients rather than unweighted binary indicators.

3.3 Phase 2: Core Innovation — Confidence-Aware `fuse()` Operator

Let $X_{\text{proj}} \in \mathbb{R}^{B \times K_h \times M \times d_h}$ denote the multi-head projected feature tensor, where B is the batch size, K_h is the number of attention heads, M is the number of modalities, and d_h is the feature dimension per head. Let $C \in [0,1]^{B \times M}$ denote the dynamic confidence tensor, and $\Omega \in \{0,1\}^{B \times M}$ denote the binary observation mask.

The generalized `fuse()` operator executes three mathematical operations:

Step 1: Multi-Head Tensor Broadcasting

The 2D confidence tensor is unsqueezed and broadcast across the attention-head dimension, gated by the observation mask:

$$\tilde{C}_{i,k,m} = C_{i,m} \cdot \Omega_{i,m} \quad \forall k \in \{1, \ldots, K_h\} \tag{3.3}$$

Step 2: ϵ-Clamped Normalization

To guarantee numerical stability against division-by-zero when an entire view is unobserved, an ϵ-clamp ($\epsilon = 10^{-9}$) is applied across modality summation:

$$S_{i,k} = \max\left(\sum_{m=1}^{M} \tilde{C}_{i,k,m},\ 10^{-9} \right) \tag{3.4}$$

$$\alpha_{i,k,m} = \frac{\tilde{C}_{i,k,m}}{S_{i,k}} \tag{3.5}$$

Step 3: Confidence-Gated Aggregated Projection

The fused representation for patient i at attention head k is calculated as the reliability-weighted inner product:

$$Z_{i,k} = \sum_{m=1}^{M} \alpha_{i,k,m} \cdot X_{\text{proj}(i,k,m)} \tag{3.6}$$

The final patient embedding $Z_i \in \mathbb{R}^{K_h \cdot d_h}$ concatenates representations across all K_h attention heads.

3.4 Algorithmic Implementation

Listing 3.1 provides the complete PyTorch implementation of the generalized `fuse()` operator.

```python
import torch
import torch.nn as nn

def fuse(x_proj, mask, mode="original", att_lin=None, external_confidence=None):
    """
    Generalized Confidence-Aware Multi-Head Attention Fusion Operator.

    Args:
        x_proj: Tensor of shape [B, num_heads, num_modalities, head_dim]
        mask: Tensor of shape [B, num_modalities], 1 if present/imputed, 0 if missing
        mode: "original" (parametric attention) or "external" (confidence-driven)
        att_lin: Linear projection layer for original mode
        external_confidence: Tensor of shape [B, num_modalities] containing values in [0, 1]

    Returns:
        fused_embeddings: Tensor of shape [B, num_heads, head_dim]
        att_weights: Attention distribution tensor across modalities
    """
    batch_size, num_heads, num_modalities, head_dim = x_proj.shape

    if mode == "external":
        if external_confidence is None:
            raise ValueError("mode='external' requires an external_confidence tensor.")

        # 1. Expand 2D tensor across multi-head dimension: [B, 1, M] -> [B, H, M]
        ext_conf = external_confidence.unsqueeze(1).expand(-1, num_heads, -1)
        expanded_mask = mask.unsqueeze(1).expand(-1, num_heads, -1)

        # 2. Mask out genuinely missing views and normalize with epsilon clamp
        ext_conf = ext_conf * expanded_mask
        sum_conf = torch.sum(ext_conf, dim=-1, keepdim=True).clamp(min=1e-9)
        att_weights = ext_conf / sum_conf

    elif mode == "original":
        # Parametric dot-product self-attention baseline
        scores = att_lin(x_proj).squeeze(-1)  # [B, H, M]
        scores[mask.unsqueeze(1).expand(-1, num_heads, -1) == 0] = -1e9
        att_weights = torch.softmax(scores, dim=-1)
    else:
        raise NotImplementedError(f"Unsupported fusion mode: {mode}")

    # 3. Modality-axis aggregated fusion: [B, H, M, 1] * [B, H, M, D] -> [B, H, D]
    fused_embeddings = torch.sum(att_weights.unsqueeze(-1) * x_proj, dim=2)

    return fused_embeddings, att_weights
```

Listing 3.1: Algorithmic implementation of the confidence-aware `fuse()` operator.

3.5 Phase 3 & 4: Patient Graph Refinement and GraphSAGE Aggregation

3.5.1 Patient Graph Edge Formulation

To prevent synthetic artifacts from creating false patient graph topologies, edges between patient nodes u and v are restricted by a biological overlap condition:

$$\mathcal{E} = \left\{ (u, v) \;\middle|\; \sum_{m=1}^{M} (\Omega_{u,m} \cdot \Omega_{v,m}) \geq 1 \right\} \tag{3.7}$$

This rule guarantees that edges exist **only if patients share at least one verified biological modality**. Imputed profiles are strictly restricted to enriching node attribute vectors and feature-wise edge cosine similarities e'_{uv}, preventing hallucinated graph linkages.

3.5.2 GraphSAGE Neighborhood Aggregation

Downstream spatial message passing is conducted over two stacked GraphSAGE layers:

$$h_{\mathcal{N}(u)}^{(l)} = \text{AGGREGATE}\left(\left\{ h_v^{(l-1)}, \forall v \in \mathcal{N}(u) \right\} \right) \tag{3.8}$$

$$h_u^{(l)} = \text{ReLU}\left(W^{(l)} \cdot \left[h_u^{(l-1)} \parallel h_{\mathcal{N}(u)}^{(l)} \right] \right) \tag{3.9}$$

where $\parallel$ denotes concatenation and $\mathcal{N}(u)$ is the neighborhood of patient node u. The final contextual node representation $Z^{(L)}$ integrates both intrinsic multi-omics features and global cohort manifold structure.

3.6 Phase 5: Joint Multi-Task Optimization Objective

The framework is optimized end-to-end using a dual-objective loss function:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}}(y, \hat{y}) + \lambda \mathcal{L}_{\text{KL}}(P \parallel Q) \tag{3.10}$$

1. **Supervised Task Cross-Entropy Loss ($\mathcal{L}_{\text{CE}}$):** Drives accurate phenotypic cancer subtype classification:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{K} y_{i,c} \log(\hat{y}_{i,c}) \tag{3.11}$$

   where $\hat{y} = \text{Softmax}(W_c Z^{(L)} + b_c)$.

2. **Manifold Alignment Loss ($\mathcal{L}_{\text{KL}}$):** Aligns the fused latent embedding space (Q) with the raw similarity distribution (P) via Kullback-Leibler divergence:

$$\mathcal{L}_{\text{KL}}(P \parallel Q) = \sum_{i \neq j} p_{ij} \log\left(\frac{p_{ij}}{q_{ij}} \right) \tag{3.12}$$

   where p_{ij} and q_{ij} denote pairwise affinity probabilities. A scaling coefficient of $\lambda = 0.1$ balances the gradient updates.

Chapter 4

Experimental Setup and Implementation

4.1 Dataset Characteristics and Modality Missingness Patterns

The framework was evaluated across three distinct cancer cohorts from The Cancer Genome Atlas (TCGA), as detailed in Table 4.1.

Table 4.1: Multi-omics dataset missingness distributions and feature statistics.

Cohort	Total Patients	DNA Methylation Missing (%) [Present]	mRNA Missing (%) [Present]	miRNA Missing (%) [Present]	DNA Feat. Dim (d_1)
TCGA-BRCA	956	628 (65.69%) [328]	0 (0.00%) [956]	372 (38.91%) [584]	2,000
TCGA-BLCA	433	2 (0.46%) [431]	10 (2.31%) [423]	7 (1.62%) [426]	2,000
TCGA-OV	360	4 (1.11%) [356]	176 (48.89%) [184]	58 (16.11%) [302]	2,000

- **TCGA-BRCA (Breast Invasive Carcinoma):** Characterized by severe DNA methylation missingness (65.69% missing), serving as an ideal testbed for cross-modal reconstruction.
- **TCGA-BLCA (Bladder Urothelial Carcinoma):** A near-complete cohort (< 2.5% missingness across all modalities), serving as a high-sensitivity benchmark for stability and noise injection.
- **TCGA-OV (Ovarian Serous Cystadenocarcinoma):** Exhibits substantial transcriptomics missingness (48.89% missing mRNA).

4.2 Adversarial Stress-Testing Framework

To rigorously evaluate model stability under degraded and adversarial conditions, three experimental regimes were designed:

1. **Unperturbed Control:** Standard stratified TCGA evaluation split. Imputations are derived via the trained cross-modal reconstructor, with dynamic confidence assigned using validation cosine similarity.
2. **Stress Test A (False Low Confidence):** For a randomly sampled 20% of patients, confidence for a valid, present modality is artificially reduced to $C_{i,m} \approx 0.1$. This regime tests whether the architecture degrades gracefully when confidence is erroneously attenuated.

3. **Stress Test B (Noise + Falsely High Confidence):** For a randomly sampled 20% of patients, authentic biological features are replaced with Gaussian noise $\sim \mathcal{N}(0, 1)$, while confidence is spoofed to $C_{i,m} \approx 0.9$. This simulates corrupted clinical sequencing paired with faulty quality assurance.

4.3 Computational and GPU Memory Optimizations

Initial baseline executions encountered notable computational bottlenecks:

- **I/O Synchronization Latency:** Per-epoch terminal logging across Google Colab environments triggered blocking GPU-CPU synchronization stalls.
- **Broadcasting Overhead:** Python-level iterative loops across multi-head projections increased per-epoch latency. Baseline runs required ≈ 20 minutes per dataset.

To address these bottlenecks, three key optimizations were implemented:

1. **Vectorized Multi-Head Broadcasting:** Replaced iterative loops with vectorized PyTorch tensor expansions (`ext_conf.unsqueeze(1).expand(-1, H, -1)`).
2. **Persistent VRAM Pinning:** Pinned graph adjacency matrices and patient feature tensors directly in GPU memory to eliminate recurring bus transfers.
3. **I/O Throttling:** Restricted evaluation and loss logging intervals to every 10 epochs.

Together, these optimizations reduced execution runtime to **10–15 minutes per dataset**, achieving an overall **30% to 40% speedup**.

Reproducibility and Code Availability

The software pipeline was developed using Python 3.10, PyTorch 2.0.1+cu118, PyTorch Geometric 2.3.1, and scikit-learn 1.3.0. Experiments were executed on an NVIDIA Tesla V100 GPU (16GB VRAM) with CUDA 11.8 drivers. The source code, trained model checkpoints, and run scripts are archived and available for academic replication upon request.

Chapter 5

Results and Discussion

5.1 Upstream Imputation Diagnostics

The performance of the `MiRNATODNAPredictor` MLP was evaluated across validation folds using cosine similarity:

$$\text{CosSim}(Y, \hat{Y}) = \frac{Y \cdot \hat{Y}}{\|Y\|_2 \, \|\hat{Y}\|_2} \tag{5.1}$$

Table 5.1: Upstream reconstruction performance metrics on held-out validation folds.

Cohort	Mean Validation Cosine Similarity	Reliability Assessment
TCGA-BRCA	0.9417	Highly Reliable
TCGA-BLCA	0.9358	Highly Reliable
TCGA-OV	0.9034	Robust

The high cosine similarity (≥ 0.9034) confirms that cross-omics regulatory interactions provide sufficient signal to reliably reconstruct missing DNA methylation features.

5.2 Empirical Classification Benchmarks (TCGA-BRCA)

TCGA-BRCA classification results ($N = 956$) across 5-fold cross-validation are presented in Table 5.2.

Table 5.2: Empirical classification benchmarking on TCGA-BRCA ($N = 956$).

Framework	Experimental Condition	Accuracy ($\mu \pm \sigma$)	Macro-F1 ($\mu \pm \sigma$)
Original MAGNET	Unperturbed Control	0.8990 ± 0.0205	0.8806 ± 0.0200
Original MAGNET	Stress Test A (False Low Conf)	0.8917 ± 0.0172	0.8725 ± 0.0165
Original MAGNET	Stress Test B (Noise + High Conf)	0.8688 ± 0.0185	0.8440 ± 0.0191
Proposed Framework	**Unperturbed Control**	$\mathbf{0.9010 \pm 0.0147}$	$\mathbf{0.8803 \pm 0.0177}$
Proposed Framework	**Stress Test A (False Low Conf)**	$\mathbf{0.8948 \pm 0.0129}$	$\mathbf{0.8743 \pm 0.0169}$
Proposed Framework	**Stress Test B (Noise + High Conf)**	$\mathbf{0.8708 \pm 0.0185}$	$\mathbf{0.8460 \pm 0.0147}$

- **Variance Reduction:** The standard deviation on control classification accuracy drops from ± 0.0205 to ± 0.0147, representing a **28.3% reduction in variance**.

- **Graceful Degradation:** Macro-F1 remains consistently bounded above 0.8460 across both stress regimes.

5.3 Empirical Classification Benchmarks (TCGA-BLCA)

The near-complete TCGA-BLCA cohort ($N = 433$) highlights the framework's resistance to synthetic noise injection, as detailed in Table 5.3.

Table 5.3: Empirical classification benchmarking on TCGA-BLCA ($N = 433$).

Framework	Experimental Condition	Accuracy ($\mu \pm \sigma$)	Macro-F1 ($\mu \pm \sigma$)
Original MAGNET	Unperturbed Control	0.9682 ± 0.0133	0.7941 ± 0.0915
Original MAGNET	Stress Test A (False Low Conf)	0.9682 ± 0.0133	0.7941 ± 0.0915
Original MAGNET	Stress Test B (Noise + High Conf)	0.9591 ± 0.0245	0.7084 ± 0.1882
Proposed Framework	**Unperturbed Control**	$\mathbf{0.9705 \pm 0.0116}$	$\mathbf{0.8185 \pm 0.0629}$
Proposed Framework	**Stress Test A (False Low Conf)**	$\mathbf{0.9705 \pm 0.0116}$	$\mathbf{0.8185 \pm 0.0629}$
Proposed Framework	**Stress Test B (Noise + High Conf)**	$\mathbf{0.9682 \pm 0.0085}$	$\mathbf{0.7988 \pm 0.0359}$

- **Baseline Performance Gain:** Baseline Macro-F1 increases from 0.7941 to 0.8185 ($+0.0244$ gain), accompanied by a **31.3% reduction in standard deviation** (± 0.0629 vs. ± 0.0915).
- **Catastrophic Failure Neutralization:** Under Stress Test B, baseline MAGNET collapses to a Macro-F1 of 0.7084 ± 0.1882. In contrast, the proposed framework preserves a Macro-F1 of 0.7988 ± 0.0359, delivering an absolute advantage of $+0.0904$ (112.76% relative ratio) and maintaining high stability.

5.4 Robustness, Variance Diagnostics, and Condition Spread

Model stability across conditions was quantified using the Coefficient of Variation (CV $= \sigma/\mu$) and the Condition Spread ($\Delta_{\text{max}-\text{min}} = \text{Metric}_{\text{max}} - \text{Metric}_{\text{min}}$), as shown in Table 5.4.

Table 5.4: Relative variability (Coefficient of Variation) and condition performance spread.

Metric / Evaluation Domain	Baseline MAGNET	Proposed Framework	Instability Reduction
BLCA Macro-F1 CV (Control)	0.1152	0.0768	**33.3% Reduction**
BLCA Macro-F1 CV (Stress B)	0.2657	0.0449	**83.1% Reduction**
BRCA Accuracy CV (Control)	0.0228	0.0163	**28.5% Reduction**
BLCA Accuracy Spread (Δ)	0.0091	0.0023	**–74.7% Spread**
BLCA Macro-F1 Spread (Δ)	0.0857	0.0197	**–77.0% Spread**
BRCA Macro-F1 Spread (Δ)	0.0366	0.0343	**–6.3% Spread**

5.5 Analytical Takeaways and Discussion

The empirical advantages of the proposed framework stem from two core mechanisms:

1. **Neutralizing Noise Propagation:** Baseline MAGNET uses a binary mask that treats valid observations and noisy features identically. When corrupted noise is introduced, it receives standard multi-head attention weighting, propagating corrupted embeddings across patient graph edges. Our confidence gating bounds the influence of degraded inputs, preserving local manifold structure during GraphSAGE aggregation.

2. **Effective High-Missingness Imputation:** In TCGA-BRCA, where 65.69% of patients lack DNA methylation data, the `MiRNATODNAPredictor` reliably reconstructs missing epigenetic signals (cosine similarity $= 0.9417$). When coupled with confidence scaling, the downstream classifier retains cohort-wide predictive signal without the variance inflation typical of unweighted imputation.

Chapter 6

Internship Work, Engineering Hurdles, and Learning

6.1 Division of Collaborative Responsibilities

To maintain clear academic attribution, project responsibilities were structured as follows:

- **Collaborative Contributions (Gautam Anand & Gunjan Gupta):**
  - Ingestion of raw genomic, epigenomic, and transcriptomic cohorts from TCGA.
  - Cohort partitioning into clean complete-case pools and missing modality pools.
  - Exploratory replication and hyperparameter inspection of baseline MAGNET.
  - Preliminary cross-modal MLP reconstructor prototyping (miRNA $\rightarrow$ DNA).
  - Implementation of initial validation cosine similarity calculation routines.
- **Independent Contributions (Yash Prakash Gupta):**
  - Conceptualization, mathematical derivation, and PyTorch implementation of the generalized confidence-aware `fuse()` operator.
  - Multi-head tensor unsqueezing, broadcasting, and ϵ-clamped normalization logic.
  - Downstream 2-layer GraphSAGE message-passing pipeline integration.
  - Formulation of the patient edge refinement rule requiring ≥ 1 verified biological modality.
  - Design and execution of adversarial stress-testing protocols (Stress Tests A and B).
  - System-level GPU memory profiling, persistent VRAM tensor pinning, and vectorization optimizations.

6.2 Engineering Challenges Overcome

1. **Tensor Dimension Mismatches:**
   - *Problem:* Injecting a 2D confidence tensor $C \in \mathbb{R}^{B \times M}$ into a 4D multi-head projected tensor $X_{\text{proj}} \in \mathbb{R}^{B \times K_h \times M \times d_h}$ caused PyTorch dimension broadcast mismatches.
   - *Solution:* Derived explicit multi-head tensor unsqueezing (`unsqueeze(1).expand(-1, num_heads, -1)`), maintaining broadcast alignment without allocating redundant memory buffers.

2. **Division-by-Zero and NaN Gradient Explosions:**
    - *Problem:* When a patient missed multiple modalities, unconstrained summation in attention normalizers yielded near-zero denominators, causing division-by-zero errors and NaN gradients.
    - *Solution:* Implemented an ϵ-clamped normalization step (clamp$(\text{min} = 10^{-9})$), ensuring numerical stability during backpropagation.
3. **Colab Notebook I/O Synchronization Stalls:**
    - *Problem:* Printing per-epoch training metrics over cloud-hosted notebooks triggered recurring CPU-GPU CUDA synchronization barriers, doubling runtimes.
    - *Solution:* Vectorized the validation graph construction pipeline and throttled logging to 10-epoch intervals, cutting runtimes by 30–40%.

6.3 Technical Competencies Acquired

- **Advanced Deep Learning & Tensor Engineering:** Vectorized multi-dimensional PyTorch tensor manipulations, custom autograd attention operators, and multi-task loss balancing.
- **Geometric Deep Learning & GNNs:** Practical implementation of Graph Convolutional Networks, Graph Attention Networks, and inductive GraphSAGE using PyTorch Geometric (PyG).
- **Bioinformatics & Oncology Data Workflows:** Processing and normalizing TCGA datasets across diverse molecular platforms (Illumina Infinium 450k Human Methylation, RNA-Seq v2, microRNA-Seq).
- **System Profiling & Robustness Auditing:** Designing adversarial stress-testing protocols to systematically evaluate edge cases and failure modes in deep architectures.

Chapter 7

Conclusion and Future Work

7.1 Summary of Research Contributions

This research resolves the modality missingness problem in multimodal cancer subtyping without resorting to cohort pruning or unconstrained naive imputation:

1. **Confidence-Aware Attention Fusion:** Formulated and implemented the `fuse()` operator, allowing continuous empirical reliability scores to dynamically gate multi-head attention representations.
2. **End-to-End Robust Architecture:** Built a modular 5-phase pipeline connecting cross-modal neural reconstruction (miRNA $\rightarrow$ DNA) to a GraphSAGE graph classifier protected by strict biological overlap constraints.
3. **Empirical Performance and Stability:** Improved TCGA-BLCA baseline Macro-F1 from 0.7941 to 0.8185 (a 31.3% reduction in variance), while demonstrating strong resistance to noise injection (outperforming baseline MAGNET by $+0.0904$ in Macro-F1 under Stress Test B).

7.2 Limitations of the Current System

- **Unidirectional Cross-Modal Estimation:** The reconstructor currently maps only miRNA $\rightarrow$ DNA, omitting reverse estimation or multi-source synthesis.
- **Cohort-Level Confidence Metric:** Imputation confidence is derived from validation-set cosine similarity across the cohort rather than per-patient uncertainty bounds.
- **Discrete Edge Sparsity:** Graph edges require at least one verified biological modality, which can limit connectivity for samples with high missingness.

7.3 Future Research Directions

1. **Evidential Uncertainty Estimation:** Integrate Bayesian Neural Networks or Monte Carlo Dropout into the reconstructor to generate sample-specific predictive distributions σ_i^2 for per-patient confidence gating.
2. **Bidirectional Multi-Omics Variational Autoencoders (VAEs):** Develop cross-modal VAEs capable of cyclic translation across arbitrary missing patterns ($M \rightarrow N$).

3. **Continuous Soft Graph Topologies:** Leverage high-confidence imputed embeddings directly within continuous similarity edge metrics, relaxing rigid discrete overlap constraints while preserving manifold structure.

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